

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Industrial and Systems Engineering

2. **Subject Name:** Statistical Learning with Applications
L-T-P: 3-1-0 Credits: 4 credit

3. **Pre-requisites:**
Operations Research I (IM21201)
Operations Research II (IM21204)
Probability and Statistics (MA20205)

4. **Syllabus and reference books:**

Syllabus:

Introduction Statistical Learning, Linear Regression, Classification, Resampling, Model Selection and Regularization, Non-linear Regression, Tree-based Methods, Artificial Neural Network, Support Vector Machines, Unsupervised Learning

Text Books:

1) James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An Introduction to Statistical Learning: With Applications in R, New York: springer.

Reference Books:

- 1) Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning: data mining, inference, and prediction. Springer Science & Business Media.
- 2) Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). Mining of massive data sets. Cambridge university press.
- 3) Haykin, S. S. (2009). Neural networks and learning machines/Simon Haykin.
- 4) Bishop, C. M. (2006). Pattern Recognition and Machine learning. Springer.

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Introduction Statistical Learning	2
2.	Linear Regression	4
3.	Classification	8
4.	Resampling Methods	4
5.	Model Selection and Regularization	4
6.	Non-linear Regression	6
7.	Tree-Based Methods	8
8.	Statistical Learning using ANN	6

9.	Support Vector Machines	4
10.	Unsupervised Learning	6
Total number of hours		52